EIGHTH ANNUAL ECC
UNDERGRADUATE
MATHEMATICS COMPETITION

April 9, 2005

Time control: Three hours

NO BOOKS, NOTES, CALCULATORS, COMPUTERS OR NON-TEAM-MEMBERS may be consulted.

PLEASE BEGIN EACH PROBLEM ON A NEW SHEET OF PAPER. Team identification and problem number should be clearly given at the top of each sheet of paper submitted.

Each problem counts 10 points. Partial credit for significant but incomplete work. For full credit, answers must be fully justified. But this may in some cases simply mean showing all work and reasoning. Have fun!

1. Sum of digits.

Let $n = 999\ldots999$ be the integer which in decimal form is represented by a string of 2005 9s. Find the sum of the digits of n^2. Defend your answer.

2. Limit of a sequence.

Find

$$\lim_{n\to\infty} n(\sqrt{n^2+1} - n).$$

Justify your answer.

3. Find a formula.

The positive integers are partitioned into finite sets as follows:

$$\{1\},\ \{2,3\},\ \{4,5,6\},\ \{7,8,9,10\},\ \text{etc.}$$

Thus, there is one integer in the first set, two in the second, three in the third, and so on. Let $g(n)$ be the sum of the n integers in the n-th subset. Thus, $g(1) = 1$, $g(2) = 5$, $g(3) = 15$, etc. Find, with proof, a general formula for $g(n)$.

4. Integer solutions.

For what integral values of m do there exist integer solutions of the following system of equations?

$$13x + 11y = 700$$
$$y = mx - 1$$

(Justify your answer.)

5. Integral equation.

Solve for x:

$$\int_1^x \frac{dt}{1 + \lfloor \log_{10} t \rfloor} = 2005.$$

Here, as usual, $\lfloor u \rfloor$ denotes the greatest integer less than or equal to u.

6. It's an integer.

Let r and s be the roots of the equation $x^2 - 9x + 1 = 0$. Prove that

$$r^{2005} + s^{2005}$$

is an integer.

7. An inequality.

Let a and b be positive real numbers. Prove that

$$\frac{(a+b)^3}{a^2 b} \geq \frac{27}{4},$$

and determine when equality occurs.

8. Minimum area ratio.

The rectangles S and T in the (x, y)-plane have sides parallel to the coordinate axes. Rectangle S lies inside or on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, and the ellipse lies inside or on T. Find, with proof, the smallest possible value of the ratio

$$r = \frac{\text{Area}(T)}{\text{Area}(S)}.$$

9. Probability it's divisible by 10.

A random number generator randomly generates integers from the set $\{1, 2, \ldots, 9\}$ with equal probability. Find the probability that after n numbers are generated, their product is a multiple of 10. Defend your answer.

10. Radical equation.

Find the (positive) root of the following equation and prove that there is only one positive root:

$$x = \sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + x}}}}.$$